APM1220: Applied Multivariate Data Analysis
FA 1

Problem 2.2

Given

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}.$$

(a) $5A$

$$5A = 5\begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}.$$

Therefore,

$$\boxed{5A = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}}$$

(b) BA

The dimensions are

$$B_{3\times 2}A_{2\times 2},$$

so the multiplication is defined and the result is a 3×2 matrix.

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}.$$

$$BA = \begin{bmatrix} 4(-1) + (-3)(4) & 4(3) + (-3)(2) \\ 1(-1) + (-2)(4) & 1(3) + (-2)(2) \\ (-2)(-1) + 0(4) & (-2)(3) + 0(2) \end{bmatrix}$$

$$= \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}.$$

Therefore,

$$\boxed{BA = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}}$$

(c) $A'B'$

First, take the transpose of each matrix:

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix}, \qquad B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}.$$

Therefore,

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}.$$

$$A'B' = \begin{bmatrix} (-1)(4) + 4(-3) & (-1)(1) + 4(-2) & (-1)(-2) + 4(0) \\ 3(4) + 2(-3) & 3(1) + 2(-2) & 3(-2) + 2(0) \end{bmatrix}$$

$$= \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}.$$

Therefore,

$$\boxed{A'B' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}}$$

(d) $C'B$

First,

$$C' = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix}.$$

Therefore,

$$C'B = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}.$$

$$C'B = \begin{bmatrix} 5(4) + (-4)(1) + 2(-2) & 5(-3) + (-4)(-2) + 2(0) \end{bmatrix}.$$

$$= \begin{bmatrix} 12 & -7 \end{bmatrix}.$$

Therefore,

$$\boxed{C'B = \begin{bmatrix} 12 & -7 \end{bmatrix}}$$

(e) Is AB defined?

The dimensions are

$$A_{2\times 2} \qquad \text{and} \qquad B_{3\times 2}.$$

For AB to be defined, the number of columns of A must equal the number of rows of B. Here,

$$2 \neq 3.$$

Therefore,

$$\boxed{AB \text{ is not defined.}}$$

Problem 2.4

Suppose A^{-1} and B^{-1} exist.

(a) Prove that $(A')^{-1} = (A^{-1})'$

Since A^{-1} exists,

$$AA^{-1} = I.$$

Taking the transpose of both sides gives

$$(AA^{-1})' = I'.$$

Using the properties

$$(XY)' = Y'X'$$

and

$$I' = I,$$

we obtain

$$(A^{-1})'A' = I.$$

Similarly, since

$$A^{-1}A = I,$$

taking the transpose gives

$$(A^{-1}A)' = I',$$

so

$$A'(A^{-1})' = I.$$

Therefore, $(A^{-1})'$ is both a left and right inverse of A'. Hence,

$$\boxed{(A')^{-1} = (A^{-1})'}.$$

(b) Prove that $(AB)^{-1} = B^{-1}A^{-1}$

Consider

$$(B^{-1}A^{-1})(AB).$$

Using associativity,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B.$$

Since

$$A^{-1}A = I,$$

we have

$$B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

We also check the other order:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}.$$

Since

$$BB^{-1} = I,$$

we obtain

$$AIA^{-1} = AA^{-1} = I.$$

Thus, $B^{-1}A^{-1}$ is the inverse of AB. Therefore,

$$\boxed{(AB)^{-1} = B^{-1}A^{-1}}.$$

Problem 2.8

Given

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix},$$

find the eigenvalues λ_1 and λ_2, the associated normalized eigenvectors e_1 and e_2, and the spectral decomposition of A.

Step 1: Find the eigenvalues

The eigenvalues satisfy

$$\det(A - \lambda I) = 0.$$

We have

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}.$$

Therefore,

$$\det(A - \lambda I) = (1-\lambda)(-2-\lambda) - 4.$$

Expanding,

$$(1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6.$$

Thus,

$$\lambda^2 + \lambda - 6 = 0.$$

Factoring,

$$(\lambda - 2)(\lambda + 3) = 0.$$

Therefore,

$$\boxed{\lambda_1 = 2, \qquad \lambda_2 = -3}.$$

Step 2: Find the eigenvector associated with $\lambda_1 = 2$

Solve

$$(A - 2I)e_1 = 0.$$

We have

$$A - 2I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}.$$

Let

$$e_1 = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$-x + 2y = 0,$$

so

$$x = 2y.$$

Choose $y = 1$. Then $x = 2$, giving the eigenvector

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Its norm is

$$\sqrt{2^2 + 1^2} = \sqrt{5}.$$

Therefore, the normalized eigenvector is

$$\boxed{e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}}.$$

Step 3: Find the eigenvector associated with $\lambda_2 = -3$

Solve

$$(A + 3I)e_2 = 0.$$

We have

$$A + 3I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$

Let

$$e_2 = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$4x + 2y = 0,$$

which gives

$$y = -2x.$$

Choose $x = 1$. Then $y = -2$, giving

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Its norm is

$$\sqrt{1^2 + (-2)^2} = \sqrt{5}.$$

Therefore,

$$\boxed{e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}}.$$

Step 4: Spectral decomposition

For a symmetric matrix, the spectral decomposition is

$$A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'.$$

First,

$$e_1 e_1' = \frac{1}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$

Similarly,

$$e_2 e_2' = \frac{1}{5} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}.$$

Therefore,

$$A = 2 \left(\frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \right) - 3 \left(\frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \right).$$

Hence,

$$\boxed{A = 2e_1 e_1' - 3e_2 e_2'}.$$

Problem 2.11

Let $A = \{a_{ij}\}$ be a $p \times p$ diagonal matrix, where

$$a_{ij} = 0, \qquad i \neq j.$$

Thus,

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{pp} \end{bmatrix}.$$

Expanding the determinant along the first row,

$$|A| = a_{11} A_{11} + 0 + \cdots + 0 = a_{11} A_{11}.$$

The submatrix corresponding to A_{11}, obtained by deleting the first row and first column, is also diagonal:

$$A_{11} = \begin{bmatrix} a_{22} & 0 & \cdots & 0 \\ 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{pp} \end{bmatrix}.$$

Expanding its determinant along the first row gives

$$A_{11} = a_{22} A_{22}.$$

Therefore,

$$|A| = a_{11} a_{22} A_{22}.$$

Repeating this process for each successive submatrix gives

$$|A| = a_{11} a_{22} a_{33} \cdots a_{pp}.$$

Hence,

$$\boxed{|A| = a_{11} a_{22} \cdots a_{pp}}.$$

Problem 2.24

Given

$$\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) Find Σ^{-1}

Since Σ is diagonal, its inverse is obtained by taking the reciprocal of each nonzero diagonal element:

$$\boxed{\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

(b) Eigenvalues and eigenvectors of Σ

Because Σ is diagonal, its eigenvalues are its diagonal elements:

$$\boxed{\lambda_1 = 4, \qquad \lambda_2 = 9, \qquad \lambda_3 = 1}.$$

For $\lambda_1 = 4$,

$$\boxed{e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}.$$

For $\lambda_2 = 9$,

$$\boxed{e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}.$$

For $\lambda_3 = 1$,

$$\boxed{e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}.$$

Thus,

$$\boxed{(\lambda_1, \lambda_2, \lambda_3) = (4, 9, 1)}$$

with corresponding eigenvectors

$$\boxed{e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.}$$

(c) Eigenvalues and eigenvectors of Σ^{-1}

We have

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore, its eigenvalues are

$$\boxed{\lambda_1 = \frac{1}{4}, \qquad \lambda_2 = \frac{1}{9}, \qquad \lambda_3 = 1}.$$

The corresponding eigenvectors remain the same:

$$\boxed{e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.}$$

Problem 2.30

Given

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}, \qquad \mu_X = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix},$$

and

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}.$$

Partition X as

$$X = \begin{bmatrix} X^{(1)} \\ X^{(2)} \end{bmatrix},$$

where

$$X^{(1)} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \qquad X^{(2)} = \begin{bmatrix} X_3 \\ X_4 \end{bmatrix}.$$

Also,

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}.$$

(a) $E(X^{(1)})$

The first two elements of the mean vector are

$$\boxed{E(X^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}}.$$

(b) $E(AX^{(1)})$

Using

$$E(AX^{(1)}) = AE(X^{(1)}),$$

we have

$$E(AX^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Therefore,

$$E(AX^{(1)}) = 4 + 6 = 10.$$

Hence,

$$\boxed{E(AX^{(1)}) = 10}.$$

(c) $\mathrm{Cov}(X^{(1)})$

The covariance matrix for $X^{(1)}$ is the upper-left 2×2 block of Σ_X:

$$\boxed{\mathrm{Cov}(X^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}.$$

(d) $\mathrm{Cov}(AX^{(1)})$

Using

$$\mathrm{Cov}(AX^{(1)}) = A\,\mathrm{Cov}(X^{(1)})A',$$

we obtain

$$\mathrm{Cov}(AX^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

First,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix}.$$

Thus,

$$\mathrm{Cov}(AX^{(1)}) = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 + 4 = 7.$$

Therefore,

$$\boxed{\mathrm{Cov}(AX^{(1)}) = 7}.$$

(e) $E(X^{(2)})$

The last two elements of the mean vector are

$$\boxed{E(X^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}}.$$

(f) $E(BX^{(2)})$

Using

$$E(BX^{(2)}) = BE(X^{(2)}),$$

we obtain

$$E(BX^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Therefore,

$$E(BX^{(2)}) = \begin{bmatrix} 2-2 \\ 4-1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Hence,

$$\boxed{E(BX^{(2)}) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}}.$$

(g) $\mathrm{Cov}(X^{(2)})$

The covariance matrix for $X^{(2)}$ is the lower-right 2×2 block of Σ_X:

$$\boxed{\mathrm{Cov}(X^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}}.$$

(h) $\mathrm{Cov}(BX^{(2)})$

Using

$$\mathrm{Cov}(BX^{(2)}) = B\,\mathrm{Cov}(X^{(2)})B',$$

we have

$$\mathrm{Cov}(BX^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

First,

$$B\,\mathrm{Cov}(X^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}.$$

Therefore,

$$\mathrm{Cov}(BX^{(2)}) = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

Thus,

$$\mathrm{Cov}(BX^{(2)}) = \begin{bmatrix} 13+20 & 26+10 \\ 20+16 & 40+8 \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}.$$

Hence,

$$\boxed{\mathrm{Cov}(BX^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}}.$$

(i) $\mathrm{Cov}(X^{(1)}, X^{(2)})$

The covariance between $X^{(1)}$ and $X^{(2)}$ is the upper-right 2×2 block of Σ_X:

$$\boxed{\mathrm{Cov}(X^{(1)}, X^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}}.$$

(j) $\mathrm{Cov}(AX^{(1)}, BX^{(2)})$

Using

$$\mathrm{Cov}(AX^{(1)}, BX^{(2)}) = A\,\mathrm{Cov}(X^{(1)}, X^{(2)})B',$$

we obtain

$$\mathrm{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

First,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 \end{bmatrix}.$$

Therefore,

$$\mathrm{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

Thus,

$$\mathrm{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 4-4 & 8-2 \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}.$$

Hence,

$$\boxed{\mathrm{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}}.$$

Problem 2.41

Given

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}, \qquad \mu_X = \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix},$$

and

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Also,

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}.$$

(a) Find $E(AX)$

Using

$$E(AX) = AE(X) = A\mu_X,$$

we obtain

$$E(AX) = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}.$$

Calculating each element,

$$E(AX) = \begin{bmatrix} 1(3) - 1(2) + 0(-2) + 0(0) \\ 1(3) + 1(2) - 2(-2) + 0(0) \\ 1(3) + 1(2) + 1(-2) - 3(0) \end{bmatrix}.$$

Therefore,

$$E(AX) = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}.$$

Hence,

$$\boxed{E(AX) = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}}.$$

(b) Find $\mathrm{Cov}(AX)$

For a linear transformation,

$$\mathrm{Cov}(AX) = A\Sigma_X A'.$$

Since

$$\Sigma_X = 3I_4,$$

we have

$$\mathrm{Cov}(AX) = 3AA'.$$

First,

$$A' = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix}.$$

Therefore,

$$AA' = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix}.$$

Calculating the entries,

$$AA' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

Thus,

$$\mathrm{Cov}(AX) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

and therefore

$$\boxed{\mathrm{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}}.$$

The variances are

$$\boxed{\text{Var}(Y_1) = 6, \qquad \text{Var}(Y_2) = 18, \qquad \text{Var}(Y_3) = 36,}$$

where

$$Y = AX.$$

The covariances are

$$\boxed{\text{Cov}(Y_1, Y_2) = 0,}$$

$$\boxed{\text{Cov}(Y_1, Y_3) = 0,}$$

and

$$\boxed{\text{Cov}(Y_2, Y_3) = 0.}$$

(c) Which pairs of linear combinations have zero covariances?

The three linear combinations are

$$Y_1 = X_1 - X_2,$$

$$Y_2 = X_1 + X_2 - 2X_3,$$

and

$$Y_3 = X_1 + X_2 + X_3 - 3X_4.$$

From part (b),

$$\text{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}.$$

The off-diagonal entries are all zero. Therefore,

$$\boxed{\text{Cov}(Y_1, Y_2) = 0,}$$

$$\boxed{\text{Cov}(Y_1, Y_3) = 0,}$$

and

$$\boxed{\text{Cov}(Y_2, Y_3) = 0.}$$

Thus, every pair of the three linear combinations has zero covariance.